
Anti psychotic drugs

Anti psychotics

- Anti psychotics are drugs that primarily affect psyche (mental process) & useful in psychiatric disorders
- Also called
 - ❑ Psychopharmacological agents
 - ❑ Psychotropic drugs
 - ❑ Neuroleptics
 - ❑ Major tranquillizers

Schizophrenia

- Geek words: *Skhizo*(to split) & *Phern* (mind) w/c means the split b/n the emotions & the intellect
- It is one particular kind of psychosis

Main clinical features

Positive symptoms : delusions, hallucinations & thought disorder

Commonly occur in acute phase of the illness

Usually respond to antipsychotic drug therapy

Negative symptoms : apathy, social withdrawal & lack of drive

Occur in the chronic phase of illness

Tend to be resistant to drug therapy

Etiology & Pathogenesis

1. Genetic & environmental factors

- Number of susceptibility genes are identified w/c show strong but incomplete hereditary tendency
- Some environmental factors have been identified as possible predisposing factors (Eg. maternal virus infections)

2. Neurochemical theories

- A change in amine NTs especially DA has been proposed as a cause of psychosis
- The main neurochemical theories centre on DA & glutamate
- However, 5-HT & other NTs also might also involve

2.1. Dopamine theory

The theory suggests that schizophrenia is caused by abnormality of DA receptors in particular D₂

Evidences to proof the theory

- Drugs like
 - Levodopa (DA precursor)
 - Amphetamine (DA release)
 - Apomorphine & bromocriptine (DA agonists)
- either aggravate or produce psychosis de novo in some patients**

- Most antipsychotic drug act by blocking D_2 on mesolimbic system
- There is an $\uparrow$ in DA receptor density in treated & untreated schizophrenics when compared with normal control
- DA receptor density has been found to be $\uparrow$ ed in the brains of schizophrenics who have not been treated with antipsychotic drugs (postmortem)
- After successful treatment with antipsychotics, there is change in the amount of **homovanillic acid** in the CSF, plasma & urine

Evidences against DA theory

- Drugs w/c block DA should bring complete cure
.....practically not true
- Phencyclin (NMDA receptor antagonist) produce much more schizophrenia like symptom
- Atypical antipsychotics has less affinity for D₂

2.2 . Glutamate theory

- NMDA receptor antagonists (phencyclidine, ketamine & dizocilpine) produce psychotic symptoms
- Reduced glutamate concentrations & glutamate receptor densities have been reported in postmortem schizophrenic brains

2.3. Other theories

- Many effective antipsychotic drugs, in addition to blocking DA receptors also act as 5-HT_{2A} receptor antagonists
- 5-HT modulates dopamine pathways
- Whether 5-HT_{2A} receptor blockade accounts directly for their antipsychotic effects, or merely reduces undesirable side effects associated with D₂-receptor antagonists remains controversial.

3. Anatomic damage

In psychotics brain, there are certain areas w/c gets atrophied

However recent studies do n't support this theory

Classification of antipsychotic drugs (based on chemical structure)

1. Phenthiazine derivatives

1.1. With an aliphatic side chain

Chlorpromazine (CPZ), trifluoperazine

1.2. With a piperidine ring side chain

Thioridazine, mesoridazine, piperacetazine

1.3. With a piperazine ring side chain

Fluphenazine, trifluoperazine

2. Butyrophenones

Haloperidole, trifluoperidole, penfluperidole

3. Thioxanthenes Chlorprothixene, flupenthixol

3. Atypical neuroleptics Clozapine,
risperidone, olanzapine, ziprasidone,
aripiprazole

3. Miscellaneous
Reserpine

Broad category of antipschotic drugs

1st generation ('typical') & 2nd generation ('atypical')

No clear distinction b/n them ; however, grouping is based on

- ❑ Receptor profile
- ❑ Incidence of extrapyramidal side effects (less in atypical group)
- ❑ Efficacy in 'treatment-resistant' patients
- ❑ Efficacy against negative symptoms

(2nd generation) Newer generation causes metabolic syndrome

Eg. Clozapine (atypical group)

- ❑ Has broad spectrum of activity than traditional antipsychotics but less affinity for D_2
- ❑ Has some efficacy for treatment resistant schizophrenia & negative symptoms
- "atypical" is used to describe antipsychotic drugs w/c do n't cause EPS
- While 'typical' antipsychotic drugs associated with anticholinergic, sedation, & cardiovascular side effects in addition to EPS

Depot antipsychotics

- Esterification of the antipsychotic with long chain fatty acid
- The drug will be released at constant rate for long time
- Reduce compliance problem
- However reduced flexibility of dosage, pain at site of administration, high incidence of EPS & weight gain

Some antipsychotic available as Depot

Haloperidol, Flupenthixol, Zuclopenthixol,
Fluphenazine, pipothiazine

Mechanism of action

- All effective antipsychotic drugs block D_2 receptors
- The degree of blockade to other actions on different receptors considerably varies

Chlorpromazine: $\alpha_1 > 5\text{-HT}_{2A} > \mathbf{D}_2 > D_1$

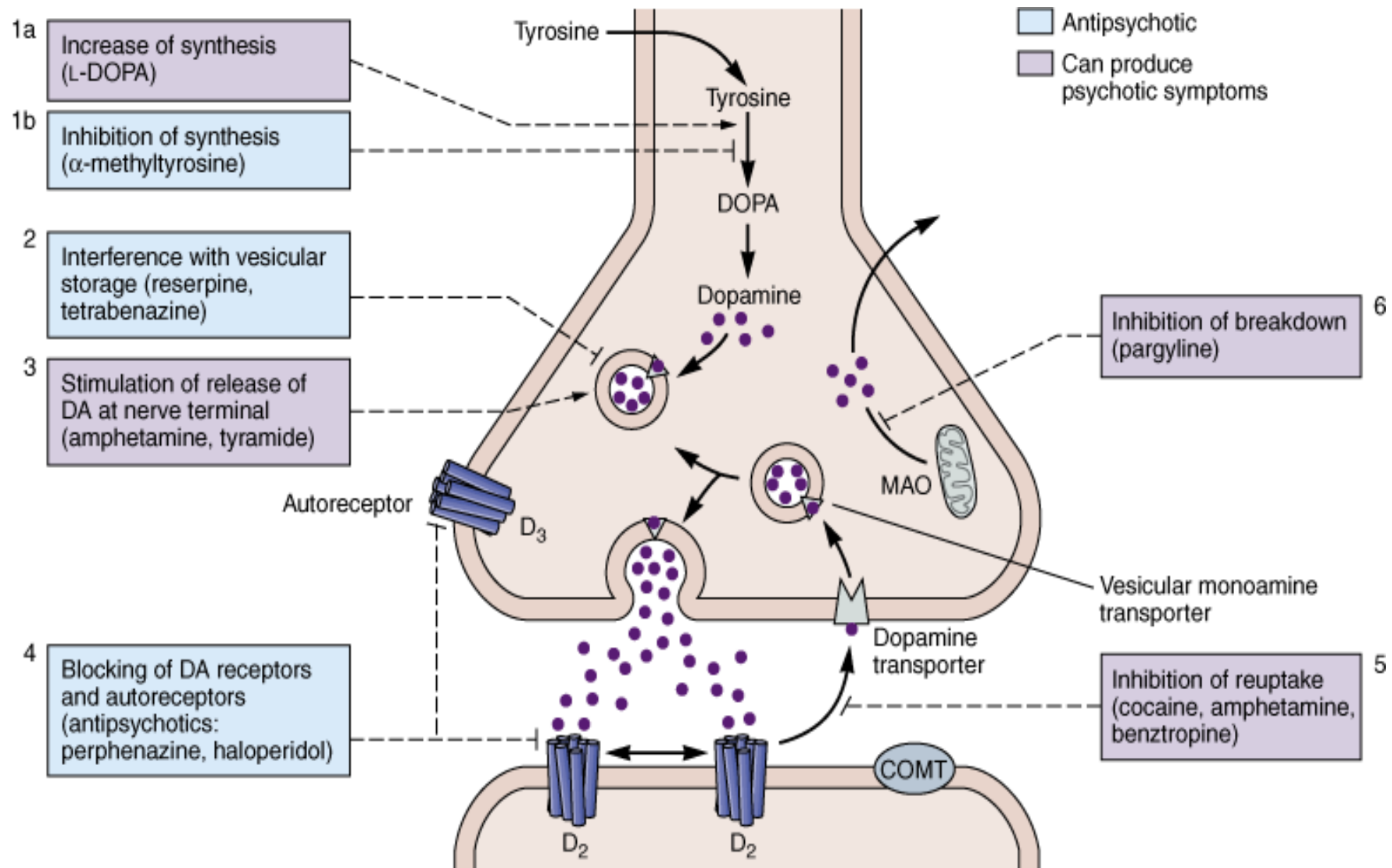
Haloperidole: $\mathbf{D}_2 > \alpha_1 > D_4 > 5\text{-HT}_{2A} > D_1 > H_1$

Clozapine: $D_4 = \alpha_1 > 5\text{-HT}_{2A} > D_1 = \mathbf{D}_2$

Aripiprazole: $\mathbf{D}_2 = 5\text{-HT}_{2A} > D_4 > \alpha_1 = H_1 \gg D_1$

Olanzapine: $5\text{-HT}_{2A} > H_1 > D_4 > \mathbf{D}_2 > \alpha_1 = H_1 \gg D_1$

- Antipsychotic action has shown good correlation with the capacity to bind to D₂ receptor
- There no clear correlation with antipsychotic activity & the capacity to bind with D₁, D₃, & D₄
- Activities on other NT receptors may determine side effect profile



Pharmacological action

1. ANS

- Varying degree of α adrenergic blocking activity
- More potent drugs have lesser α blocking activity
- Anticholinergic property is generally weak

2. Local anesthetic

CPZ is potent LA as procaine however it has irritation

3. CVS

Antipsychotics produce postural hypotension by central & peripheral action on sympathomimetic tone

4. Endocrine effects

Increase prolactine secretion by blocking the inhibitory effect of DA

- ❑ Galactorrhoea, gynaecomastia
- ❑ increased libido (women), decreased libido (men)

Use of antipsychotics

1. Schizophrenia

2. Anxiety

- Should not be used for simple anxiety b/c of autonomic & EPS
- BZD_s are preferable; however, those not responding or having psychiatric base for the anxiety may be treated with neuroleptics

3. As antiemetics

Control wide range of drug & disease induced vomiting at dose much lower than those needed for psychosis but ineffective in motion sickness

4. Other uses

- Potentiate hypnotics, analgesics, & anesthetics
- Intractable hiccough may respond to parenteral CPZ
- In tetanus, CPZ is secondary drug to achieve skeletal muscle relaxation

Adverse effects

I. Dose related toxicity

1. CNS

- Drowsiness, lethargy, mental confusion
- Increased appetite & weight gain (not with haloperidol)
- Aggravation of seizures in epileptics
 - Non epileptics may develop seizure at high dose of some antipsychotics such as clozapine & olanzapine

Potent phenothiazines such as ziprasidone have little effect on seizure threshold

2. CVS

- ❑ Postural hypotension
- ❑ Palpitation

3. Anticholinergic

- Dry mouth, blurred vision

3. Endocrine

- Hyperprolactinemia
 - Atypical antipsychotics do n't raise prolactin level
-

5. Extrapyramidal disturbances

a. Pseudo parkinsonism

- Rigidity, tremor, hypokinesia, mask like face
- Appears b/n 1- 4 weeks of therapy & persist unless dose is reduced
 - Anticholinergic anti PD drugs can be given together with antipsychotics

b. Acute muscular dystonias

- Muscle spasm mostly in linguo-facial muscles
- Grimacing , tongue thrusting, torticollis, locked jaw

- More common in children below 10 & girls
- Occurs within a few hrs of single dose or at 1st week of therapy
- i.m injection of central anticholinergic (promethazine or hydroxyzine) can clear the reaction within 10 – 15 minutes

c. Akathisia

- Restlessness, feeling discomfort, agitation as complete desire to move about without anxiety

- Occurs within 1-8 weeks of therapy & misleads as 'exacerbation of psychosis'
- No specific antidote available
 - Central anticholinergic may reduce the intensity but propranolol is more effective
 - Addition of diazepam may also be used
 - However, most cases of akathisia require dose reduction or alternative antipsychotic

d. Malignant neuroleptic syndrome

- Rarely occurs with potent antipsychotics
 - Patient develops marked rigidity, immobility, tremor, fever, semi consciousness, fluctuating BP & HR
 - Lasts 5 – 10 days after withdrawal & may be fatal
 - Anticholinergics are of no help rather large dose of bromocriptine may be use full
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e. Tardive dyskinesia

- Manifests as purposeless involuntary facial & limb movements like constant chewing, pouting, puffing of cheeks & lip licking
 - More common in elderly women
 - May subside months or yrs after withdrawal of the treatment or may be life long
 - No satisfactory solution found
-

II. Hypersensitivity

1. Cholestatic jaundice
2. Skin rash, urticaria, contact dermatitis , photosensitivity (more common with CPZ)
3. Agranulocytosis rarely (more common with clozapine)
4. Myocarditis

Drug interaction

1. Neuroleptics potentiate all CNS depressant
 - Hypnotics, anxiolytics, alcohol, opioids, antihistamines & analgesics
2. Neuroleptics block the action of levodopa & DA agonists in parkinsonism
3. Antihypertensive effect of clonidine & methyldopa is reduced
4. They are poor enzyme inducers

Drug of choice

- Individual patients differ in their response to different antipsychotics
- There is no way to predict w/c patient will respond better to w/c drug
- However drug selection should consider state of the patient & side effect profile of the drug

Eg. If the patient is aggressive, sedating drugs such as CPZ would be drug of choice

Haloperidol is drug of choice if postural hypotension is a problem

If there is difficulty in frequent administering of the drug go for depot antipsychotic drugs

Antidepressant drugs

Depression

- At any given moment about 3–5% of the population is depressed & an estimated 10% of people may become depressed during their lives
- ”depression” is a misleading term
- Every one in the normal course of daily life will experience mood alteration
 - In this context depression does not represent a disorder or illness

- Lowered mood as normal response to the ups & downs of living is more correctly termed as sadness & happiness
- Depression is defined as disorders of mood rather than disturbances of thought or cognition
- Major depression & dysthymia (minor) are pure depressive syndromes whereas bipolar disorder & cyclothymic disorder signify depression in association with mania

The symptoms of depression

- Emotional symptoms
 - Misery, apathy & pessimism
 - Low self-esteem: feelings of guilt, inadequacy & ugliness
 - Indecisiveness, loss of motivation.
- Biological symptoms
 - Retardation of thought & action
 - Loss of libido
 - Sleep disturbance & loss of appetite

Etiology & pathogenesis

- Genetic causes
- Environmental factors
- Biochemical factors
 - Deficiency of NT amines in certain part of the brain (NA, 5-HT & DA)
- Endocrine factors
 - ↑ plasma cortisol level in depressed patients
 - Dexamethasone suppression test

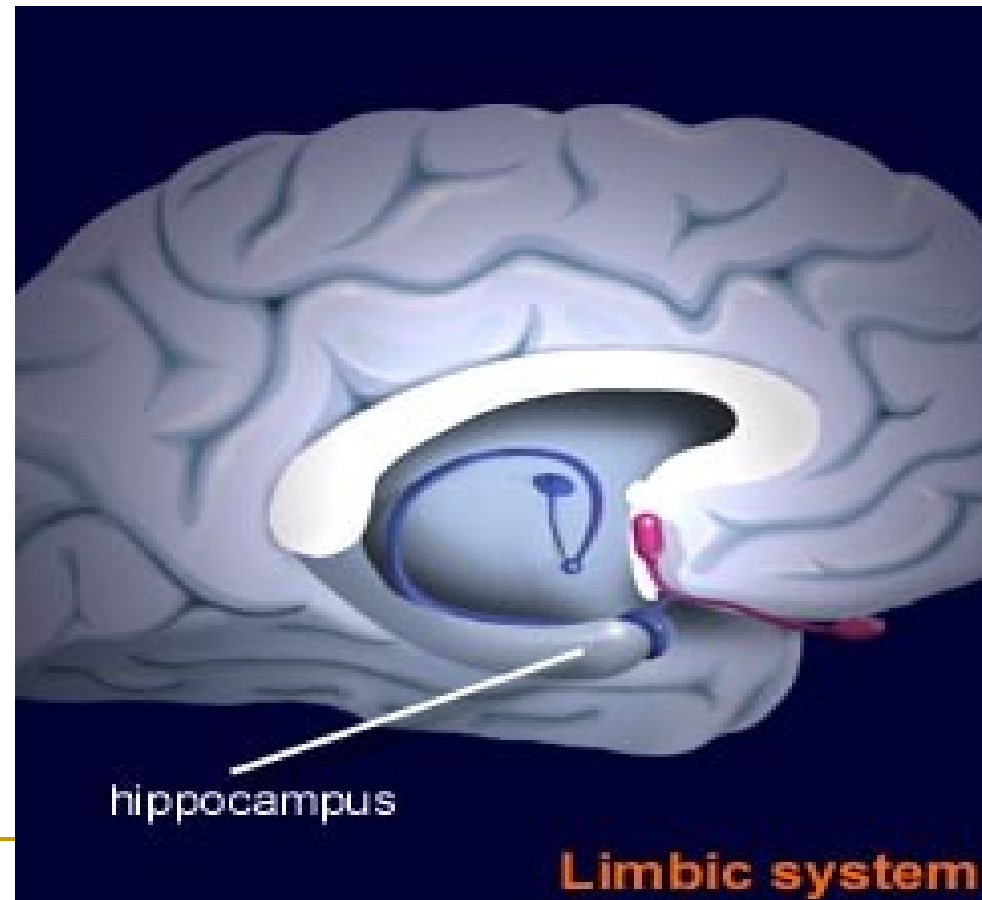
- Monoamine theory

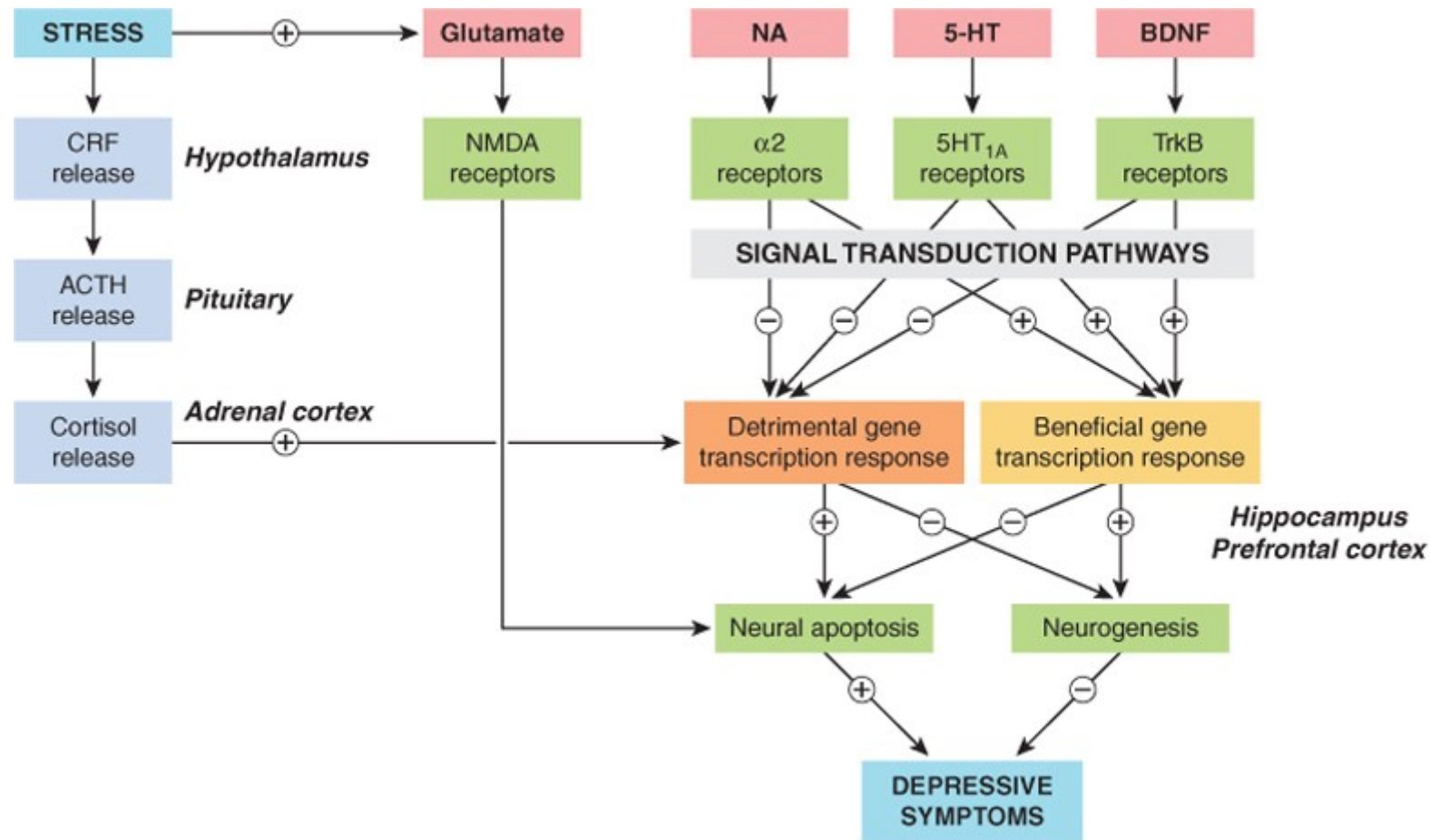
- The theory states that depression is caused by a functional deficit of monoamine transmitters (NA &/or 5-HT) at certain sites in the brain, while mania results from a functional excess

Pharmacological evidence for monoamine hypothesis

Drug(s)	Principal action	Effect in depressed Patients
TCA reuptake	Inhibit NA & 5-HT	Mood ↑
MAOI HT	Increase stores of NA & 5-HT	Mood ↑
Reserpine	Inhibits NA & 5-HT storage	Mood ↓ (calming of manic patients)
α-Methyl tyrosine	Inhibits NA synthesis	Mood ↓
Methyldopa	Inhibits NA synthesis	Mood ↑
Electro convulsive therapy	↑ CNS responses to NA & 5-HT	

- Recent evidence suggests that depression may be associated with neurodegeneration & reduced neurogenesis in the hippocampus





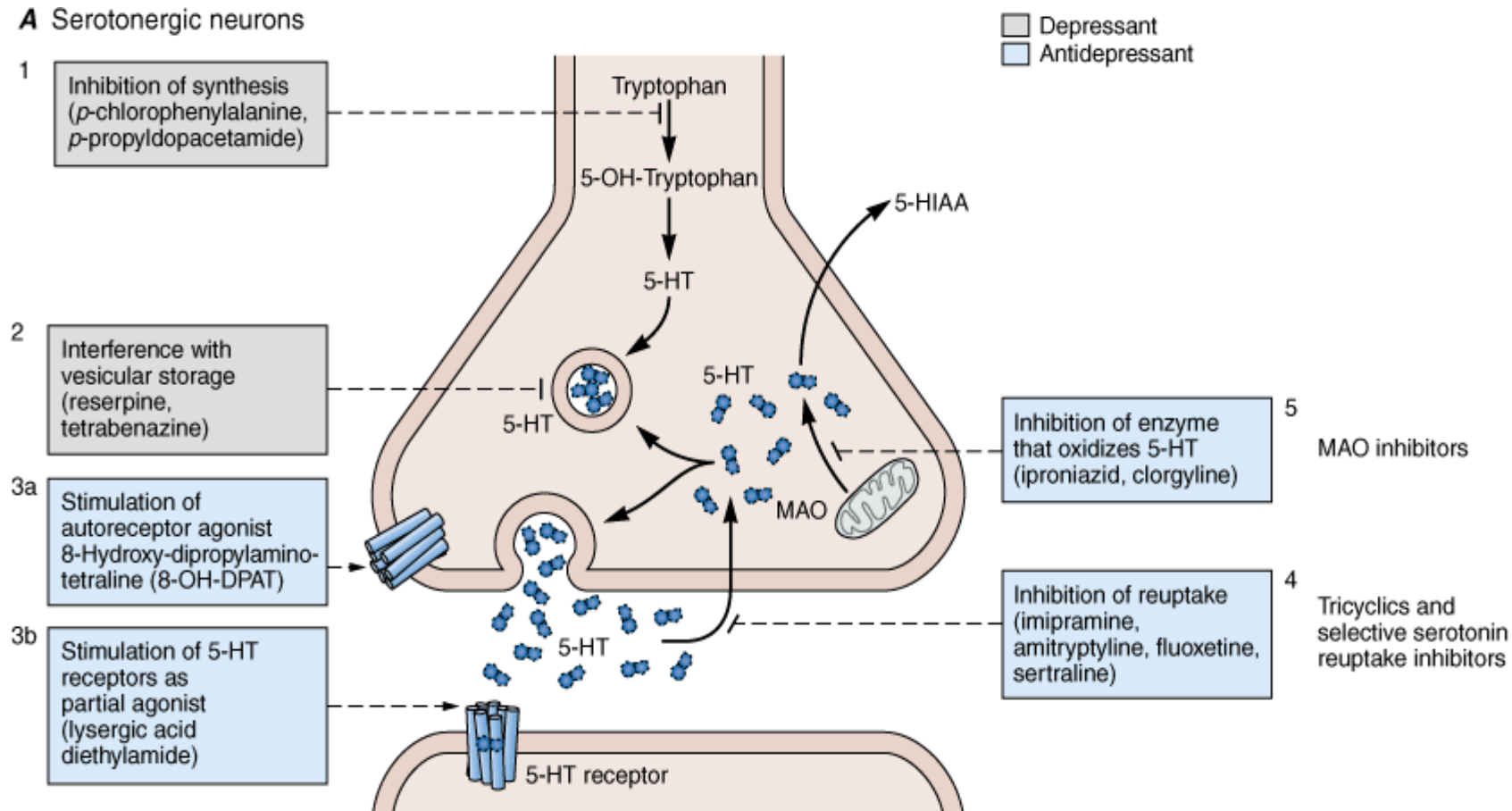
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- Prodepressive pathways involve the hypothalamic-pituitary-adrenal axis
- The antidepressive pathways involve the monoamines & the BDNF

Classification of antidepressants

1. Tricyclic antidepressants (TCAs)
2. Selective serotonin reuptake inhibitors (SSRIs)
3. Monoamine oxidase inhibitors (MAOIs)
4. Miscellaneous ('atypical') antidepressants

Mechanism of action of antidepressants

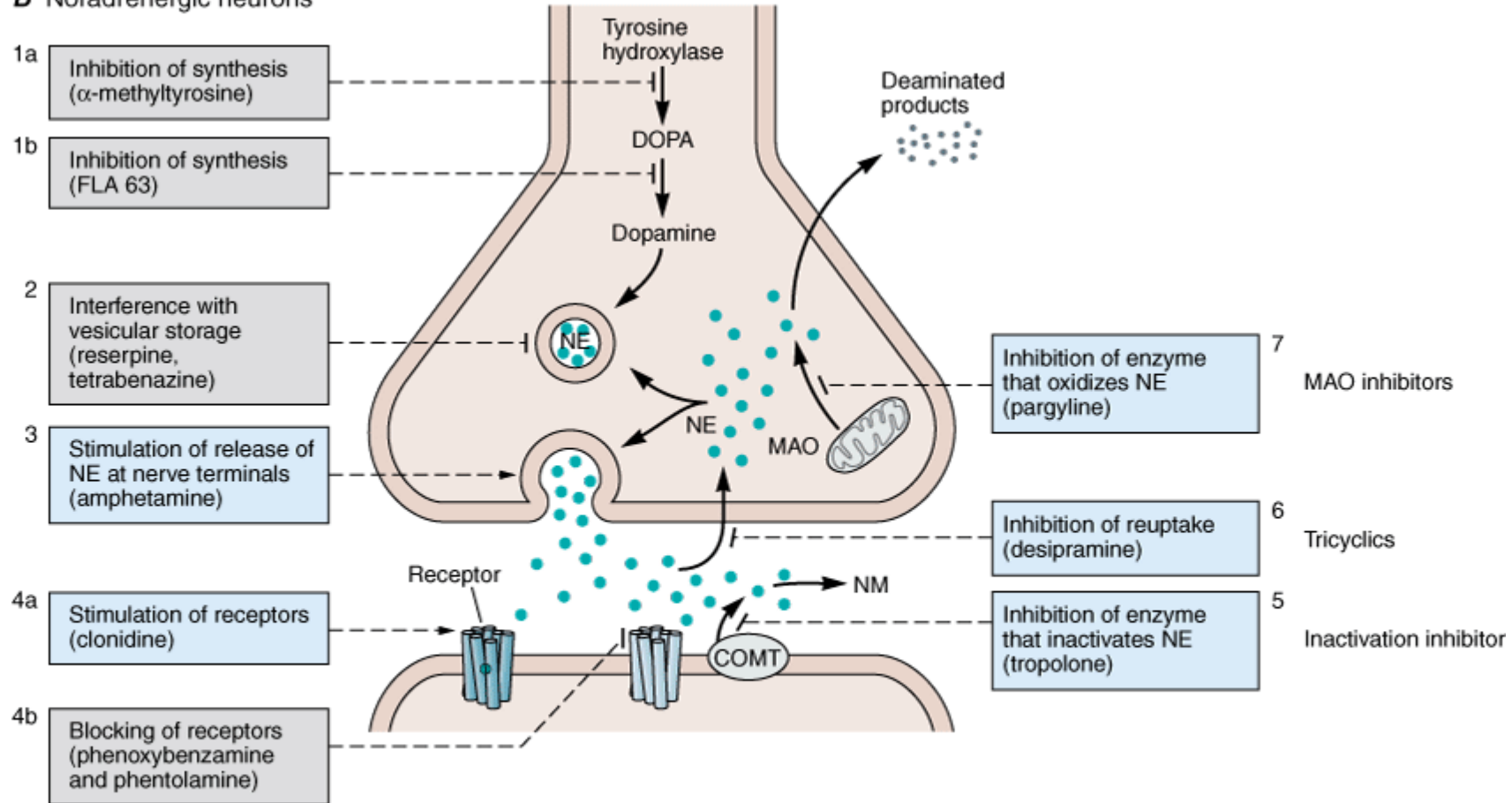


Source: Fauci AS, Kasper DL, Braunwald E, Hauser SL, Longo DL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine*, 17th Edition: <http://www.accessmedicine.com>

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Mechanism of action of antidepressants

B Noradrenergic neurons



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I. Tricyclic antidepressants (TCAs)

A. NA + 5-HT reuptake inhibitors

Imipramine

Amitriptyline

Trimipramine

Dothiepin

Doxepin

Clomipramine

B. Predominantly NA reuptake inhibitors

Amoxapine

Desipramine

Nortriptylin

Reboxetine

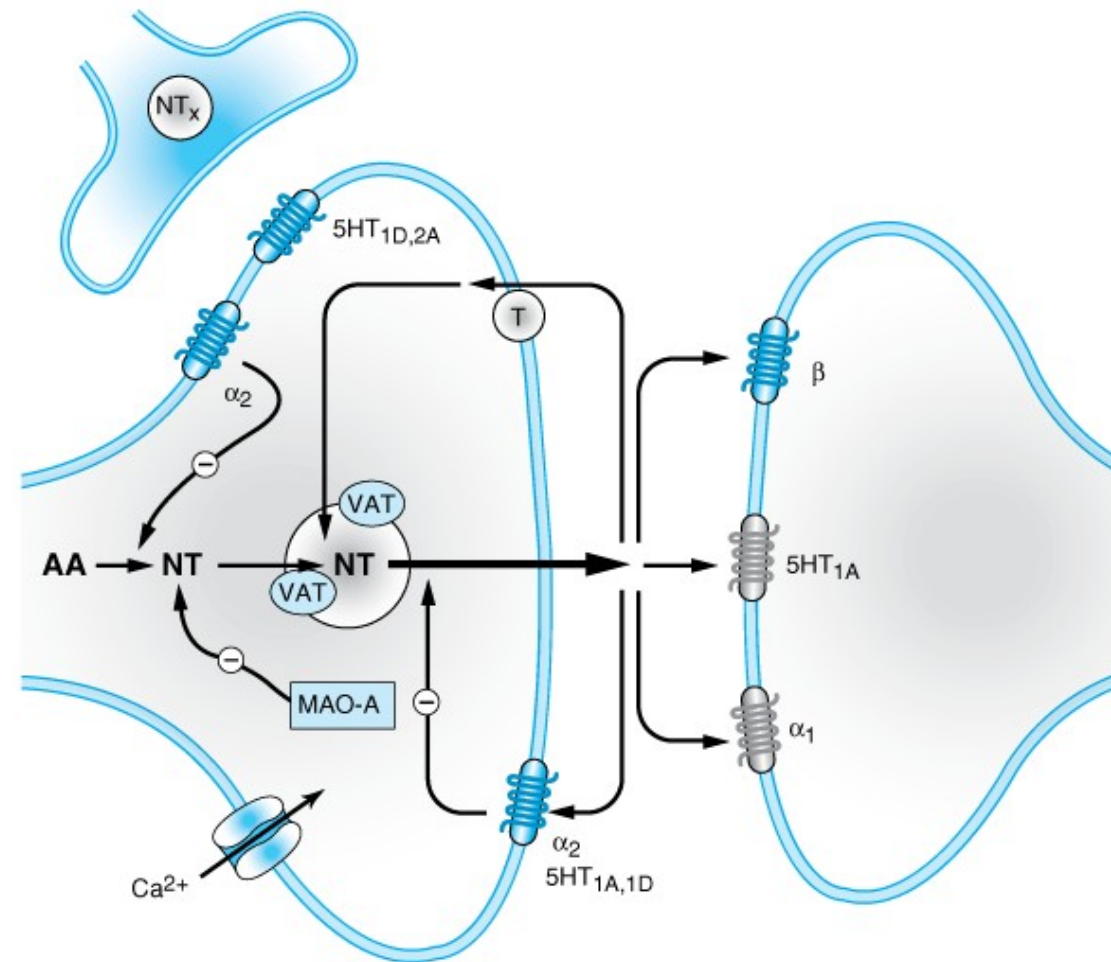
Mechanism of TCAs

- Block reuptake of NA &/or 5-HT
 - ↑NTs concentration on synaptic cleft
- On the 1st few weeks of treatment the increased NTs activate presynaptic autoreceptors (α_2 , 5-HT_{1A} & 5-HT_{1D})
 - Presynaptic autoreceptors
 - Mediated negative-feedback mechanisms
 - Respond to increased synaptic transmitter by down-regulating NTs synthesis & release

- Uptake blockage occurs quickly but antidepressant action develops after weeks
- Mianserin (atypical) has no uptake blockage action rather it block presynaptic receptors
- Trimipramine is weak NA/5-HT reuptake blocker but an equally effective antidepressant

Therefore antidepressant action of TCAs is beyond reuptake inhibition

- After long term administration of TCAs (2-3 wks)
 - The drug desensitize presynaptic auto receptors (α_2 , 5-HT_{1A} & 5-HT_{1D})
 - ↑concentration of NTs in synaptic cleft
 - Increased sensitivity of postsynaptic receptors
 - Hence, the net effect after long term therapy is enhanced noradrenergic & serotonergic transmission
- ❖ The receptor change might be more related to the antidepressant action than the reuptake inhibition mechanism



Pharmacological action

1.CNS

In normal individuals

- ❑ Clumsy feeling, tiredness, lightheadedness, sleepiness, difficulty in concentration & thinking

In depressed patients

- ❑ After 2 -3 weeks of continuous treatment mood gradually elevated & start taking interest in self & surrounding
- Drugs are not euphorants rather antidepressant
- Lower seizure threshold & produce convulsion

2.ANS

Most TCAs are potent anticholinergics

3.CVS

- ❑ Tachycardia (due to anticholinergic & NA potentiation action)
- ❑ Postural hypotension (due to cardiovascular reflex & α_1 blockade)
- ❑ ECG change & cardiac arrhythmias (T wave suppression or inversion)

Major limitations of conventional TCAs

- Anticholinergics, cardiovascular & neurological side effects
- Relatively low safety margin, dangerous in over dose
- Lag time 2-4 weeks before antidepressant action manifests
- Incomplete response by significant number of patients & some do not respond

II. Selective serotonin reuptake inhibitors (SSRIs)

Fluoxetine

Maprotiline

Fluvoxamine

Trazodone

Paroxetine

Nefazodone

Citalopram

Sertraline

SSRIs

- 1st line drugs
 - Due to their relatively safe & acceptability
- Produce little or no sedation
- Not produce anticholinergic side effects
- Devoid of α AR blocking action (no postural hypotension)- suitable for elderly patients
- No seizure precipitating propensity & not inhibit cardiac conduction
- No weight gain

-
- ❑ For TCAs & SSRIs as onset of action is slow
 - Treatment should be for at least 4-6 weeks before concluding that the drug is ineffective
 - If there is a partial response, treatment should be continued for several more weeks before increasing the dose
 - Treatment should continue for at least 4 months following remission
-

III. Monoamine oxidase inhibitors (MAOIs)

- MAO-A is primarily responsible for NE, 5-HT & tyramine metabolism
- MAO-B is more selective for DA
- Non-selective MAOIs or selective MAO-A inhibitors have antidepressant effect
- Older non selective MAOIs :Tranylcypromine, phenelzine & isocarboazid
 - Irreversible, long-acting & non-selective
- Moclobemide (MAO-A inhibitor)
 - Reversible, short acting & selective

IV. Miscellaneous ('atypical') antidepressants

Trazodone

- ❑ Selective but less efficiently block 5-HT uptake
- ❑ Prominent α blocking & 5-HT₂ antagonistic effect

Mianserin

- ❑ Not inhibit either NA or 5-HT uptake
- ❑ Block presynaptic α_2

Tianeptine

- ❑ ↑5-HT uptake, it is not sedative & stimulant
- ❑ Effective for anxiodepressive state

Amineptine

- ❑ Like tianeptine but has antidepressant action

Venlafaxine & duloxetine

‘serotonin & NA reuptake inhibitor’ = SNRI

- In contrast to older TCAs,
 - They do not interact with cholinergic, adrenergic, or histaminergic receptors
 - No sedative effect

Mirtazapine

- Block α_2 autoreceptor (on NA neuron) & heteroceptors(on 5-HT neurons)

Bupropion

- Inhibit DA & NA uptake
- Has excitant rather than sedative effect

Clinical indications of antidepressants

- Major depression
- Anxiety disorders
 - panic, generalized anxiety, social phobia & Obsessive-compulsive disorders
- Enuresis
- Chronic pain
- Migraine
 - Amitryptiline has prophylactic use

- Pruritus

- Some TCAs have antipruritic action

Eg. Topical doxepine

- Other indications

- Bulimia (fluoxetine)
 - Premenstrual dysphoric disorder (fluoxetine)
 - Attention deficit hyperkinetic disorder (imipramine, desipramine)
-

Adverse effects

■ TCAs

- ❑ Sedation, tremor & insomnia
- ❑ Blurred vision, constipation, urinary hesitancy, confusion
- ❑ Orthostatic hypotension, conduction defects, arrhythmias
- ❑ Aggravation of psychosis
- ❑ Seizures
- ❑ Weight gain, sexual disturbances

- **MAOIs**

- Headache
 - Drowsiness
 - Dry mouth
 - Weight gain
 - Postural hypotension
 - Sexual disturbances
-

- **SSRIs**

- Anxiety
 - Insomnia
 - Gastrointestinal symptoms
 - Decreased libido, sexual dysfunction
 - Teratogenic potential with paroxetine
-

Drug interactions

TCAs

1. Potentiate directly acting sympathomimetics
2. Abolish antihypertensive effect of clonidine
3. Potentiate CNS depressants
4. Phenytoin, phenylbutazone, ASA & CPZ can displace TCAs from PPB
5. PhB induce as well as competitively inhibit imipramine metabolism

6. SSRIs inhibit metabolism of several drugs
including TCAs

7. TCAs delay absorption of their own & other drugs

MAO- A inhibitors

Tyramine containing food

‘Cheese’ like reaction

Drug treatment of mania & manic depressive disorder

Drugs used

- Neuroleptics
- TCAs
- Lithium
- Carbamazepine
- Valproate
- BZDs

A. Neuroleptics

- Commonly used drugs: haloperidol or chlorpromazine or zuclopenthixol (acuphase)
- Haloperidol is drug of choice
 - As it is less sedating & free from many of cardiovascular problems of CPZ
 - Control severe behavior disturbance with additional sedatives such as lorazepam injection

-
- Atypical antipsychotics (olazapine, resperidone) & other newer antipsychotics with or without a BZD are now 1st line drugs for control of acute mania

B. Lithium carbonate

- Lithium used as carbonate salt b/c it is less hygroscopic & less gastric irritant than LiCl & other salts
- Lithium carbonate is referred to as an "antimanic" drug
- However, it is considered a "mood-stabilizing" agent
 - B/c of its primary action of preventing mood swings in patients with manic-depressive disorder

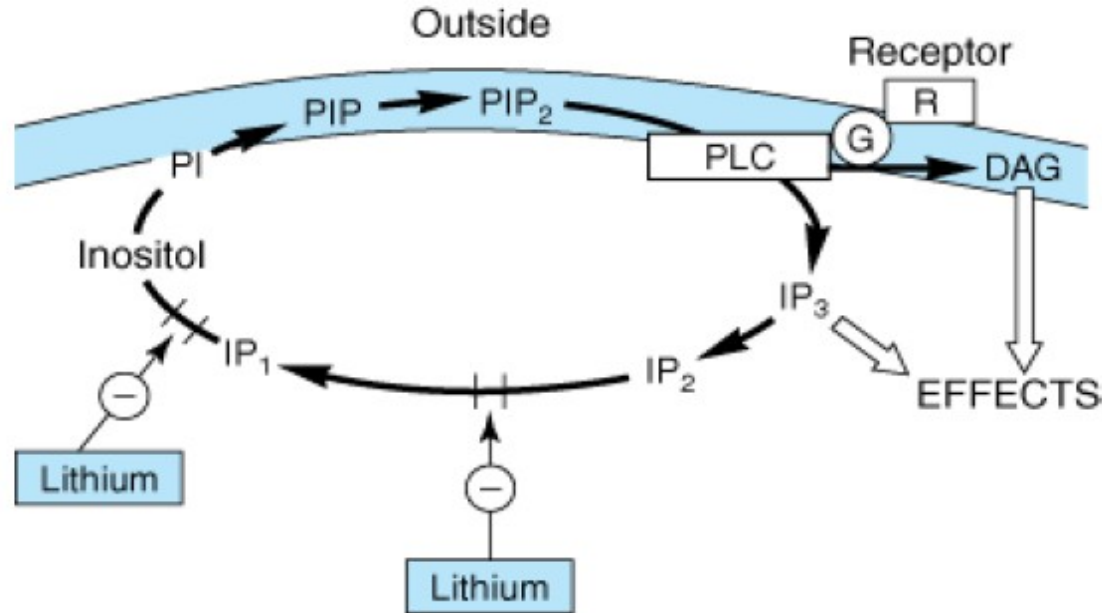
Mechanism of action

- Antimanic/mood stabilizing action remains unclear
- It has been suggested that
 1. Li^+ partly replace body Na^+ & nearly equally distributed inside & outside cells & hence it interferes ionic fluxes
 2. It ↓ NA & DA release in treated animals without affecting 5-HT release
 3. Effects on secondary messengers : one of the best-defined effects of Li^+ (action on inositol phosphates)

Effects on secondary messengers

- IP_3 & DAG : important second messengers for adrenergic & muscarinic transmission
- Blocking the pathway leads to depletion of PIP_2 (precursor of IP_3 & DAG)
- Hyperactive neurons involved in manic state may be preferentially affected
i.e. Li^+ could cause a selective depression of the overactive neurons
- Li^+ has no acute effect on normal individuals

Mechanism of action



PIP₂: phosphatidylinositol-4,5-bisphosphate

PLC: phospholipase-C

G: coupling protein

EFFECTS: activation of protein kinase C, mobilization of intracellular Ca²⁺, etc.

Pharmacokinetics

- Slowly & well absorbed (orally)
- No plasma binding & metabolism
- Gradually well distributed in total body water
- Excreted from kidney in the same way as Na^+

Clinical use of Li^+

- ❖ For acute mania

- During acute phase, Li^+ response is low & control of plasma level is difficult

- Neuroleptics (i.m. route) with or without potent BZDs (clozapam or lorazepam) is preferable in acute phase

- Li^+ treatment can be started after the episode is under control & maintenance therapy is generally for 6 – 12 months

Clinical use of Li⁺

- ❖ As prophylaxis in bipolar disorder
- ❖ For recurrent unipolar depression
 - ❑ Initially combination of TCAs with Li⁺ & then Li⁺ alone in maintenance phase
- ❖ For recurrent neuropsychiatric illness, cluster headache, & adjuvant to TCAs in resistant major depression

Adverse effects

1. Nausea, vomiting & mild diarrhea
2. Thirst & polyuria
3. CNS toxicities: coarse tremors, ataxia, motor incoordination, mental confusion, etc
 - No specific antidote for Li^+ toxicity
 - Osmotic diuretics & sodium bicarbonate infusion promote Li^+ excretion

Adverse effects

4. On long term therapy
 - Some patients develop renal diabetes insipidus
 - Goiter is also reported in few patients b/c Li^+ interfere with tyrosine iodination
5. C/I during pregnancy
 - Foetal goiter & congenital abnormalities may occur

Drug Interaction

1. Diuretics (thiazides, frusemide) promotes proximal reabsorption of Na^+ as well as Li^+ ($\uparrow$ plasma Li^+)
2. TTC, NSAIDs, & ACE inhibitors can also cause Li^+ retention
3. Li^+ enhances insulin/sulfonylurea hypoglycemia

c. TCAs

- The depressive phase of manic-depressive disorder often requires concurrent use of an antidepressant drug
 - TCAs linked to precipitation of mania
 - with more rapid cycling of mood swings but most patients do not show this effect
 - SSRIs are less likely to induce mania but may have limited efficacy
-

D. Carbamazepine

- The mode of action of carbamazepine is unclear, and oxcarbazepine is not effective
- Carbamazepine may be used to treat acute mania & also for prophylactic therapy
- May be used alone or, in refractory patients, in combination with lithium or, rarely, valproate
- The use of carbamazepine as a mood stabilizer is similar to its use as an anticonvulsant

E. Valproate

- Valproic acid is becoming recognized as an appropriate first-line treatment for mania
- It is not clear that it will be as effective as Li^+ as a maintenance treatment in all subsets of patients
- Mechanism of action of is unclear
- It shows efficacy equivalent to that of Li^+ during the early weeks of treatment
- It has been effective in some patients who have failed to respond to lithium

F. BZDs

- Used as adjuvant with other antimanic drugs
-